

Hypothesis Testing Summary

ApexPlanet Software Pvt. Ltd. - Data Analytics Internship

Objective

To determine whether there is a statistically significant difference in average sales between the Electronics and Grocery product categories.

Business Question

Do Electronics and Grocery categories have different average Total Sales?

Null Hypothesis (H_0)

There is no significant difference in the average Total Sales between the Electronics and Grocery categories.

Alternative Hypothesis (H_1)

There is a significant difference in the average Total Sales between the Electronics and Grocery categories.

Statistical Test Used

Independent Two-Sample t-test using Python (SciPy) in Google Colab.

Decision Rule

Significance level (α) = 0.05. If the p-value is less than 0.05, reject the null hypothesis; otherwise, fail to reject the null hypothesis.

Result

Insert the t-statistic and p-value obtained from your Google Colab notebook here.

Conclusion

Based on the calculated p-value, conclude whether there is a statistically significant difference between the average sales of the two categories. Update this section with your actual test result before final submission.